- Assess data on climate change to identify a problem that CCUS can address
- Use critical thinking skills, creativity, and familiarity with current research to develop potential solutions for problems once they have been identified
- Identify and address knowledge gaps
- Engage with stakeholders to identify strengths and weaknesses associated with implementing a proposed CCUS project
- Effectively communicate, with both novices and experts, about scientific, economic, and policy aspects of CCUS
- Demonstrate the ability to productively contribute to a team identifying and proposing a CCUS project

EBGN598. POLITICAL ECONOMY OF THE ENERGY TRANSITION. 3.0 Semester Hrs.
This course provides an overview of economics, business, and political topics that are commonly found in the energy transition. Many of the assignments relate back to skills that are needed to interact with economics, business, and policy professionals. The course is designed for students with little, if any, social science or business training. Students will build a basic knowledge of economics, finance, and business issues that are relevant to energy markets and industries.

SYGN598. CARBON REDUCTION: CAPTURE & UTILIZATION. 3.0 Semester Hrs.

This course provides an overview of the technologies used for decarbonization with an introduction to the chemistry of the molecule and the reactive CO2 capture technologies, carbon capture and separation technologies, thermodynamics, and practical applications from a CCUS systems development perspective.

Program Director

Yilin Fan, Petroleum Engineering, Assistant Professor

Associated Faculty

Ali Tura, Geophysics, Professor

Anuj Chauhan, Chemical and Biological Engineering , Professor

Erik Menke, Chemistry, Associate Teaching Professor

Ian Lange, Economics and Business, Associate Professor

Jennifer Miskimins, Petroleum Engineering, Professor

Manika Prasad, Geophysics, Professor

Data Science

Program Description

The Master of Data Science (non-thesis) program is designed to give candidates a foundation in statistics and computer science and also provide knowledge in a particular application domain of science or engineering. The balance between these three elements is a strength of the program and can prepare candidates for data science careers in industry, government, or for further study at the PhD level. Throughout is an emphasis on working in teams, creative problem solving, and professional development.

The Data Science Certificates are designed for college graduates and professionals interested in the emerging field of data science as applied within their individual fields of study or industries.

Primary Contact

Program Director: Soutir Bandyopadhyay
Program Co-Director: Zibo Wang
Program Coordinator: Rachel McDonald
datascience.mines.edu

Degrees Offered

- Data Science Master of Science (Non-Thesis)
- Data Science Master of Science (Non-Thesis) Online
- Data Science Graduate Certificate - Statistical Learning
- Data Science Graduate Certificate - Earth Resources
- Data Science Graduate Certificate - Petroleum Data Analytics
- Data Science Graduate Certificate - Business Analytics
- Data Science Graduate Certificate - Foundations
- Data Science Graduate Certificate - Computer Science

Data Science Master of Science (Non-Thesis)

The field of Data Science draws on elements of computer science, statistics and interdisciplinary applications to address the unique needs of gaining knowledge and insight through data analysis. This Masters Non-Thesis program is designed to give candidates a foundation in statistics and computer science and also provide knowledge in a particular application domain of science or engineering. The balance between these three elements is a strength of the program and can prepare candidates for Data Science careers in industry, government, or for further study at the PhD level. Moreover, the coursework will be flexible and tailored to each candidate. For example, the program will allow a candidate to increase his/her skills in data analytics while developing a focused area of application or alternatively allow a candidate with depth in an area of application to gain skills in statistics and computer science.

Program Requirements

Students take 18 credits of core courses, 3 credits of professional development coursework and at least 9 credits of electives for a minimum total of 30 credits.

Core Courses

Data Modeling and Statistical Learning

MATH530	INTRODUCTION TO STATISTICAL METHODS (Online Only) Students who have passed MATH335 should replace MATH530 with any of the following: MATH500, MATH531, MATH532, MATH533, MATH534, MATH535, MATH536, MATH537, MATH538, or MATH551	3.0
MATH560	INTRODUCTION TO KEY STATISTICAL LEARNING METHODS I (Online Only)	3.0

MATH561	INTRODUCTION TO KEY STATISTICAL LEARNING METHODS II (Online Only)	3.0

Machine Learning, Data Processing and Algorithms, and Parallel Computation

DSCI503	ADVANCED DATA SCIENCE	3.0
DSCI570	INTRODUCTION TO MACHINE LEARNING	3.0
DSCI575	ADVANCED MACHINE LEARNING	3.0

Professional Development Requirement

Choose courses from the table below in any combination to meet or exceed 3 credits.

Professional Development

SYGN502	INTRODUCTION TO RESEARCH ETHICS	1.0
SYGN5XX	LEADERSHIP AND TEAMWORK	1.0
SYGN683	ORAL COMMUNICATION SKILLS	1.0
SYGN684	WRITING SKILLS	2.0
LICM501	PROFESSIONAL ORAL COMMUNICATION	1.0
EBGN562	STRATEGIC DECISION MAKING	3.0
EBGN563	MANAGEMENT OF TECHNOLOGY AND INNOVATION	3.0
EBGN566	TECHNOLOGY ENTREPRENEURSHIP	3.0
EBGN577	LEADING & MANAGING HIGH PERFORMING TEAMS	3.0
PHGN503	RESPONSIBLE CONDUCT OF RESEARCH	1.0
SYGN598	SPECIAL TOPICS (Graduate Internship)	1-3

Electives

Electives are designed by the student to fulfill their interests and career goals. The intent is to give you the knowledge and skills to apply data science techniques in various fields. Although this individualized module can draw on graduate courses from across the university the courses given below are approved options:

BIOL510	BIOINFORMATICS	3.0
CBEN505	NUMERICAL METHODS IN CHEMICAL ENGINEERING	3.0
CBEN507	APPLIED MATHEMATICS IN CHEMICAL ENGINEERING	3.0
CBEN509	ADVANCED CHEMICAL ENGINEERING THERMODYNAMICS	3.0
CBEN511	NEUROSCIENCE, MEMORY, AND LEARNING	3.0
CBEN516	ADVANCED TRANSPORT PHENOMENA	3.0
CBEN518	REACTION KINETICS AND CATALYSIS	3.0
CBEN530	TRANSPORT PHENOMENA	3.0
CBEN532	TRANSPORT PHENOMENA IN BIOLOGICAL SYSTEMS	3.0
CBEN624	APPLIED STATISTICAL MECHANICS	3.0
CBEN625	MOLECULAR SIMULATION	3.0
CEEN505	NUMERICAL METHODS FOR ENGINEERS	3.0
CEEN506	FINITE ELEMENT METHODS FOR ENGINEERS	3.0
CEEN519	RISK ASSESSMENT IN GEOTECHNICAL ENGINEERING	3.0
CEEN530	ADVANCED STRUCTURAL ANALYSIS	3.0
CEEN531	STRUCTURAL DYNAMICS	3.0
CEEN533	MATRIX STRUCTURAL ANALYSIS	3.0
CEEN546	STATISTICAL METHODS FOR RELIABILITY AND ENGINEERING DESIGN	3.0
CEEN566	MICROBIAL PROCESSES, ANALYSIS AND MODELING	3.0
CHGC504	METHODS IN GEOCHEMISTRY	3.0
CHGN507	ADVANCED ANALYTICAL CHEMISTRY	3.0
CHGN583	PRINCIPLES AND APPLICATIONS OF SURFACE ANALYSIS TECHNIQUES	3.0
CSCI507	INTRODUCTION TO COMPUTER VISION	3.0
CSCI532	ROBOT ETHICS	3.0
CSCI536	HUMAN-ROBOT INTERACTION	3.0
CSCI544	ADVANCED COMPUTER GRAPHICS	3.0
CSCI555	GAME THEORY AND NETWORKS	3.0
CSCI561	THEORY OF COMPUTATION	3.0
CSCI562	APPLIED ALGORITHMS AND DATA STRUCTURES	3.0
CSCI563	PARALLEL COMPUTING FOR SCIENTISTS AND ENGINEERS	3.0
CSCI564	ADVANCED COMPUTER ARCHITECTURE	3.0
CSCI565	DISTRIBUTED SYSTEMS	3.0
CSCI571	ARTIFICIAL INTELLIGENCE	3.0
CSCI572	COMPUTER NETWORKS II	3.0
CSCI574	THEORY OF CRYPTOGRAPHY	3.0
CSCI575	ADVANCED MACHINE LEARNING	3.0
CSCI576	DEEP LEARNING	3.0
CSCI577	ADVANCED ELEMENTS OF GAMES AND GAME DEVELOPMENT	3.0
CSCI578	BIOINFORMATICS	3.0
CSCI581	QUANTUM PROGRAMMING	3.0
CSCI582	COMPUTING BEYOND CPU'S	3.0
CSCI583	IOT SECURITY AND PRIVACY	3.0
CSCI585	INFORMATION SECURITY PRIVACY	3.0
EBGN509	MATHEMATICAL ECONOMICS	3.0
EBGN525	BUSINESS ANALYTICS	3.0
EBGN526	STOCHASTIC MODELS IN MANAGEMENT SCIENCE	3.0
EBGN527	BUSINESS OPTIMIZATION MODELS	3.0
EBGN528	INDUSTRIAL SYSTEMS SIMULATION	3.0
EBGN547	FINANCIAL RISK MANAGEMENT	3.0
EBGN559	SUPPLY CHAIN ANALYTICS	3.0
EBGN560	DECISION ANALYTICS	3.0
EBGN570	ENVIRONMENTAL ECONOMICS	3.0
EBGN571	MARKETING ANALYTICS	3.0
EBGN590	ECONOMETRICS I	3.0
EBGN594	TIME-SERIES ECONOMETRICS	3.0
EBGN645	COMPUTATIONAL ECONOMICS	3.0
EENG507	INTRODUCTION TO COMPUTER VISION	3.0
EENG509	SPARSE SIGNAL PROCESSING	3.0
EENG511	CONVEX OPTIMIZATION AND ITS ENGINEERING APPLICATIONS	3.0
EENG514	DATA SCIENCE FOR ELECTRICAL ENGINEERING	3.0

EENG515	MATHEMATICAL METHODS FOR SIGNALS AND SYSTEMS	3.0
EENG519	ESTIMATION THEORY AND KALMAN FILTERING	3.0
EENG521	NUMERICAL OPTIMIZATION	3.0
EENG528	COMPUTATIONAL ELECTROMAGNETICS	3.0
EENG581	POWER SYSTEM OPERATION AND MANAGEMENT	3.0
EENG584	POWER SYSTEM RISK MANAGEMENT	3.0
GEGN532	GEOLOGICAL DATA ANALYSIS	3.0
GEGN568	POINT CLOUD DATA ANALYSIS IN EARTH SCIENCE AND ENGINEERING	3.0
GEGN579	PYTHON SCRIPTING FOR GEOGRAPHIC INFORMATION SYSTEMS	3.0
GEGN583	MATHEMATICAL MODELING OF GROUNDWATER SYSTEMS	3.0
GEGN586	NUMERICAL MODELING OF GEOCHEMICAL SYSTEMS	3.0
GEGN671	LANDSLIDES: INVESTIGATION, ANALYSIS & MITIGATION	3.0
GEOL527	SWIR (SHORT WAVELENGTH INFRA-RED) SPECTRAL ANALYSIS	1.0
GEOL613	GEOLOGIC RESERVOIR CHARACTERIZATION	3.0
GPGN658	SEISMIC WAVEFIELD IMAGING	3.0
HASS578	GLOBAL ENVIRONMENTAL ISSUES	3.0
MATH500	LINEAR VECTOR SPACES	3.0
MATH501	APPLIED ANALYSIS	3.0
MATH506	COMPLEX ANALYSIS II	3.0
MATH510	ORDINARY DIFFERENTIAL EQUATIONS AND DYNAMICAL SYSTEMS	3.0
MATH514	APPLIED MATHEMATICS I	3.0
MATH515	APPLIED MATHEMATICS II	3.0
MATH531	THEORY OF LINEAR MODELS	3.0
MATH532	SPATIAL STATISTICS	3.0
MATH533	TIME SERIES ANALYSIS AND ITS APPLICATIONS	3.0
MATH534	MATHEMATICAL STATISTICS I	3.0
MATH535	MATHEMATICAL STATISTICS II	3.0
MATH536	ADVANCED STATISTICAL MODELING	3.0
MATH537	MULTIVARIATE ANALYSIS	3.0
MATH538	STOCHASTIC MODELS	3.0
MATH540	PARALLEL SCIENTIFIC COMPUTING	3.0
MATH551	COMPUTATIONAL LINEAR ALGEBRA	3.0
MATH552	KERNEL-BASED APPROXIMATION METHODS	3.0
MATH557	INTEGRAL EQUATIONS	3.0
MATH559	ASYMPTOTICS	3.0
MATH572	MATHEMATICAL AND COMPUTATIONAL NEUROSCIENCE	3.0
MATH582	STATISTICS PRACTICUM	3.0
MEGN502	ADVANCED ENGINEERING ANALYSIS	3.0
MEGN535	MODELING AND SIMULATION OF HUMAN MOVEMENT	3.0
MEGN536	COMPUTATIONAL BIOMECHANICS	3.0
MEGN544	ROBOT MECHANICS: KINEMATICS, DYNAMICS, AND CONTROL	3.0
MEGN545	ADVANCED ROBOT CONTROL	3.0
MEGN553	COMPUTATIONAL FLUID DYNAMICS	3.0
MEGN554	ORBITAL MECHANICS	3.0
MEGN571	ADVANCED HEAT TRANSFER	3.0
MEGN586	LINEAR OPTIMIZATION	3.0
MEGN587	NONLINEAR OPTIMIZATION	3.0
MEGN588	INTEGER OPTIMIZATION	3.0
MEGN592	RISK AND RELIABILITY ENGINEERING ANALYSIS AND DESIGN	3.0
MEGN686	ADVANCED LINEAR OPTIMIZATION	3.0
MEGN688	ADVANCED INTEGER OPTIMIZATION	3.0
MNGN502	GEOSPATIAL BIG DATA ANALYTICS	3.0
PEGN501	APPLICATIONS OF NUMERICAL METHODS TO PETROLEUM ENGINEERING	3.0
PEGN513	RESERVOIR SIMULATION I	3.0
PEGN590	RESERVOIR GEOMECHANICS	3.0
PEGN624	COMPOSITIONAL MODELING - APPLICATION TO ENHANCED OIL RECOVERY	3.0
PHGN511	MATHEMATICAL PHYSICS	3.0
PHGN519	FUNDAMENTALS OF QUANTUM INFORMATION	3.0
PHGN520	QUANTUM MECHANICS I	3.0
PHGN521	QUANTUM MECHANICS II	3.0
PHGN530	STATISTICAL MECHANICS	3.0
ROBO517	INTRODUCTION TO COMPUTER VISION	3.0
ROBO550	MECHATRONICS	3.0
ROBO554	ROBOT MECHANICS: KINEMATICS, DYNAMICS, AND CONTROL	3.0
ROBO565	ADVANCED ROBOT CONTROL	3.0
ROBO572	ROBOT ETHICS	3.0
ROBO576	HUMAN-ROBOT INTERACTION	3.0

Sample Course Schedule

First Year

Fall		**lec**	**lab**	**sem.hrs**
DSCI503	ADVANCED DATA SCIENCE			3.0
DSCI570	INTRODUCTION TO MACHINE LEARNING			3.0
MATH530	INTRODUCTION TO STATISTICAL METHODS			3.0
ELECT	Elective			3.0
				12.0

Spring		**lec**	**lab**	**sem.hrs**
DSCI575	ADVANCED MACHINE LEARNING			3.0
MATH560	INTRODUCTION TO KEY STATISTICAL LEARNING METHODS I			3.0
ELECT	Elective			3.0
LICM501	PROFESSIONAL ORAL COMMUNICATION			1.0
				10.0

Second Year

Fall		lec	lab	sem.hrs
MATH561	INTRODUCTION TO KEY STATISTICAL LEARNING METHODS II			3.0
ELECT	Elective			3.0
SYGN502	INTRODUCTION TO RESEARCH ETHICS			1.0
SYGN5XX	LEADERSHIP AND TEAMWORK			1.0
				8.0

Total Semester Hrs: 30.0

Mines Combined Undergraduate / Graduate Degree Program

Students enrolled in Mines' combined undergraduate/graduate program may double count up to six credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with "B-" or better, not be substitutes for required coursework, and meet all other University, Department, and Program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

Master of Data Science (Non-Thesis) Online

The Master of Data Science (Non-Thesis) program is designed to give candidates a foundation in statistics and computer science and also provide knowledge in a particular application domain of science or engineering. The balance between these three elements is a strength of the program and can prepare candidates for Data Science careers in industry, government, or for further study at the PhD level. Throughout is an emphasis on working in teams, creative problem solving, and professional development.

This non-thesis master's program gives students a foundation in statistics and computer science, while also providing knowledge in a particular application domain of science or engineering. It is pitched a higher level of statistics and computer science than one would encounter in a typical data analytics curriculum.

The balance between these elements is a strength of the program and prepares students for data science careers in industry, government or for further study at the PhD level. The emphasis on some foundational knowledge will prepare students to be more innovative in their approach to data analysis and not rely on simply using software packages in a standard way. Moreover, the three elective courses can be tailored to each student's interests. This program allows students to either increase their skill in data analytics while *developing a focused area of application* or alternatively to allow a student with depth in one area of application to *gain skills in statistics and computer science*.

Program

Students take 18 credits of core courses, 3 credits of professional development coursework and at least 9 credits of electives for a minimum total of 30 credits.

Core Courses

MATH530	INTRODUCTION TO STATISTICAL METHODS Students who have passed MATH335 should replace MATH530 with any of the following: MATH500, MATH531, MATH532, MATH533, MATH534, MATH535, MATH536, MATH537, MATH538, or MATH551	3.0
MATH560	INTRODUCTION TO KEY STATISTICAL LEARNING METHODS I	3.0
MATH561	INTRODUCTION TO KEY STATISTICAL LEARNING METHODS II	3.0
DSCI503	ADVANCED DATA SCIENCE	3.0
DSCI570	INTRODUCTION TO MACHINE LEARNING	3.0
DSCI575	ADVANCED MACHINE LEARNING	3.0
Total Semester Hrs		**18.0**

Professional Development Requirement

Choose courses from the table below in any combination to meet or exceed 3 credits.

EBGN562	STRATEGIC DECISION MAKING	3.0
EBGN577	LEADING & MANAGING HIGH PERFORMING TEAMS	3.0
EBGN563	MANAGEMENT OF TECHNOLOGY AND INNOVATION	3.0
SYGN598	SPECIAL TOPICS (Graduate Internship)	1-3
SYGN599	INDEPENDENT STUDY	1-3

Electives

Electives are designed by the student to fulfill their interests and career goals. The intent is to give you the knowledge and skills to apply data science techniques in various fields. Although this individualized module can draw on graduate courses from across the university the courses given below are approved options:

Online Electives

AMFG523	DESIGN AND ANALYSIS OF EXPERIMENTS	3.0
CBEN524	COMPUTER-AIDED PROCESS SIMULATION	3.0
CCUS520	CLIMATE CHANGE AND SUSTAINABILITY	3.0
CSCI585	INFORMATION SECURITY PRIVACY	3.0
CSCI587	CYBER PHYSICAL SYSTEMS SECURITY	3.0
EBGN525	BUSINESS ANALYTICS	3.0
EBGN560	DECISION ANALYTICS	3.0
EBGN562	STRATEGIC DECISION MAKING	3.0
EBGN571	MARKETING ANALYTICS	3.0
EENG509	SPARSE SIGNAL PROCESSING	3.0
EENG510	ADVANCED DIGITAL SIGNAL PROCESSING	3.0
EENG515	MATHEMATICAL METHODS FOR SIGNALS AND SYSTEMS	3.0
EENG519	ESTIMATION THEORY AND KALMAN FILTERING	3.0
EENG577	ADVANCED ELECTRICAL MACHINE DYNAMICS FOR SMART-GRID SYSTEMS	3.0
EENG585	AI FOR POWER AND RENEWABLE ENERGY SYSTEMS	3.0
GEGN575	APPLICATIONS OF GEOGRAPHIC INFORMATION SYSTEMS	3.0

GEGN580	APPLIED REMOTE SENSING FOR GEOENGINEERING AND GEOSCIENCES	3.0
GEGN588	ADVANCED PLANETARY GEOGRAPHIC INFORMATION SYSTEMS	3.0
GEGN590	GIS-BASED REAL WORLD LEARNING PROJECT I - FUNDAMENTALS	3.0
GEGN592	GIS-BASED REAL WORLD LEARNING PROJECT II - ADVANCED APPLICATIONS	3.0
GEOL557	EARTH RESOURCE DATA SCIENCE 1: FUNDAMENTALS	3.0
GEOL558	EARTH RESOURCE DATA SCIENCE 2: APPLICATIONS AND MACHINE-LEARNING	3.0
GPGN545	INTRODUCTION TO DISTRIBUTED FIBER-OPTIC SENSING AND ITS APPLICATIONS	3.0
GPGN558	SEISMIC DATA INTERPRETATION AND QUANTITATIVE ANALYSIS	3.0
MNGN502	GEOSPATIAL BIG DATA ANALYTICS	3.0

Certificate Programs in Data Science

Program Requirements

There are six Certificates in Data Science. Applicants for each are required to have an undergraduate degree to be admitted into the Certificate programs. Course prerequisites, if any, are noted for each Certificate program.

Students working toward one of the Data Science Certificates are required to successfully complete 12 credits, as detailed below for each Certificate. The courses taken for the Certificates can be used towards a Master's or PhD degree at Mines, however courses used for one Data Science Certificate cannot also be counted toward another Data Science Certificate.

Graduate Certificate in Data Science - Statistical Learning (12 credits)

The Data Science - Statistical Learning Graduate Certificate is an online or residential program focusing on statistical methods for interpreting complex data sets and quantifying the uncertainty in a data analysis. The Certificate also includes gaining new skills in computer science but is grounded in statistical models for data, also termed statistical learning, rather than algorithmic approaches. Students will develop an essential skill set in statistical methods most commonly used in data science along with the understanding of the methods' strengths and weaknesses. Moreover, the coursework will cover a broad range of applications making it relevant for varied scientific and engineering domains.

Applicants must have completed the following courses, or their equivalents, with a B- or better: CSCI261 and CSCI262 Data Structures, MATH332 Linear Algebra and MATH334 Introduction to Probability.

DSCI503	ADVANCED DATA SCIENCE	3.0
DSCI530	STATISTICAL METHODS I	3.0
DSCI560	INTRODUCTION TO KEY STATISTICAL LEARNING METHODS I	3.0
DSCI561	INTRODUCTION TO KEY STATISTICAL LEARNING METHODS II	3.0

Graduate Certificate in Data Science - Earth Resources (12 credits)

The Graduate Certificate in Data Science - Earth Resources is an online program building on the foundational concepts in data science as it pertains to managing surface and subsurface Earth resources and on specific applications (use cases) from the petroleum and minerals industries as well as water resource monitoring and remote sensing of Earth change. The Certificate includes one core introductory Data Science course, two courses specific to Earth resources and one elective.

DSCI503	ADVANCED DATA SCIENCE	3.0
GEOL557	EARTH RESOURCE DATA SCIENCE 1: FUNDAMENTALS	3.0
GEOL558	EARTH RESOURCE DATA SCIENCE 2: APPLICATIONS AND MACHINE-LEARNING	3.0
ELECTIVE	(1) ELECTIVE FROM LIST BELOW	3.0

Graduate Certificate in Data Science - Earth Resources Electives (select ONE (1) from the list below):

Geospatial Focus:		
GEGN575	APPLICATIONS OF GEOGRAPHIC INFORMATION SYSTEMS	3.0
GEGN579	PYTHON SCRIPTING FOR GEOGRAPHIC INFORMATION SYSTEMS	3.0
Petroleum Focus:		
GPGN519	ADVANCED FORMATION EVALUATION	3.0
GPGN547	PHYSICS, MECHANICS, AND PETROPHYSICS OF ROCKS	3.0
GPGN558	SEISMIC DATA INTERPRETATION AND QUANTITATIVE ANALYSIS	3.0
GPGN651	ADVANCED SEISMOLOGY	3.0
PEGN522	ADVANCED WELL STIMULATION	3.0
PEGN551	PETROLEUM DATA ANALYTICS - FUNDAMENTALS	3.0
Mining Focus:		
MNGN548	INFORMATION TECHNOLOGIES FOR MINING SYSTEMS	3.0
Hydrology Focus:		
CEEN581	WATERSHED SYSTEMS MODELING	3.0
Additional Options:		
DSCI/MATH530	STATISTICAL METHODS I	3.0
EBGN525	BUSINESS ANALYTICS	3.0

Graduate Certificate in Petroleum Data Analytics (12 credits)

The Graduate Certificate in Petroleum Data Analytics is an online program building on the foundational concepts in statistics and focusing on the data foundation of the oil and gas industry, the challenges of Big Data to oilfield operations and on specific applications (use cases) for petroleum analytics. The Certificate includes two core introductory Data Science courses and two course specific to petroleum engineering.

DSCI503	ADVANCED DATA SCIENCE	3.0
DSCI530	STATISTICAL METHODS I	3.0

PEGN551	PETROLEUM DATA ANALYTICS - FUNDAMENTALS	3.0
PEGN552	PETROLEUM DATA ANALYTICS - APPLICATIONS	3.0

Graduate Certificate in Business Analytics

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The certificate is an online or residential program. The
 requirements are to complete the core course and two elective
 courses:
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Core Course

EBGN525	BUSINESS ANALYTICS	3.0

Elective Courses

EBGN559	SUPPLY CHAIN ANALYTICS	3.0
EBGN560	DECISION ANALYTICS	3.0
EBGN571	MARKETING ANALYTICS	3.0

Course substitutions may be approved on a case-by-case basis by the certificate director. Completing the certificate will also position students to apply to either the master of science in engineering and technology management degree or the master of science in data science degree, as the certificate courses can be applied to either degree.

Graduate Certificate in Data Science - Foundations (12 credits)

The Data Science - Foundations Graduate Certificate is an online or residential program focusing on the foundational concepts in statistics and computer science that support the explosion of new methods for interpreting data in its many forms. The Certificate balances an introduction to data science with teaching basic skills in applying methods in statistics and machine learning to analyze data. Students will gain a perspective on the kinds of problems that can be solved by data intensive methods and will also acquire new analysis skills outside of the certificate. Moreover, the coursework will cover a broad range of applications, making it relevant for varied scientific and engineering domains.

Applicants must have completed the following courses, or their equivalents, with a B- or better: CSCI261 and CSCI262 Data Structures, MATH332 Linear Algebra and MATH334 Introduction to Probability.

DSCI503	ADVANCED DATA SCIENCE	3.0
DSCI570	INTRODUCTION TO MACHINE LEARNING	3.0
MATH530	INTRODUCTION TO STATISTICAL METHODS	3.0
MATH560	INTRODUCTION TO KEY STATISTICAL LEARNING METHODS I	

Graduate Certificate in Data Science - Computer Science (12 credits)

The Data Science - Computer Science Graduate Certificate is an online or residential program focusing on data science concepts within computer science (e.g., computational techniques and machine learning) plus prerequisite knowledge (e.g., probability and regression). The aim of this certificate is to help students develop an essential skill set in data analytics, including (1) deriving predictive insights by applying advanced statistics, modeling, and programming skills, (2) acquiring in-depth knowledge of machine learning and computational techniques, and (3) unearthing important questions and intelligence for a range of industries, from product design to finance.

Applicants must have completed the following courses, or their equivalents, with a B- or better: CSCI261 and CSCI262 Data Structures, MATH213 Calculus III and MATH332 Linear Algebra. DSCI530 Statistical Methods I, will serve as the MATH201 Probability and Statistics prerequisite for the two machine learning courses of the certificate (DSCI570 Introduction to Machine Learning and DSCI575 Machine Learning).

DSCI503	ADVANCED DATA SCIENCE	3.0
MATH530	INTRODUCTION TO STATISTICAL METHODS	3.0
DSCI570	INTRODUCTION TO MACHINE LEARNING	3.0
DSCI575	ADVANCED MACHINE LEARNING	3.0

Courses

DSCI503. ADVANCED DATA SCIENCE. 3.0 Semester Hrs.
(I, II) This course will teach students the core skills needed for gathering, cleaning, organizing, analyzing, interpreting, and visualizing data. Students will use the python programming language and related toolkits for data manipulation and the use and application of statistical and machine learning for data analysis. The course will be primarily focused on applications, with an emphasis on working with real (non-synthetic) datasets. Students will propose and design a semester project using a dataset from their domain of interest, leveraging the concepts and skills acquired from this course (e.g., data analysis, ethical considerations, evaluation and synthesis of results, storytelling and visualization). Prerequisite: CSCI200 with a grade of C- or higher or CSCI262 with a grade of C- or higher, MATH201 or MATH334 OR Graduate level standing and at least CSCI128 or equivalent.
Course Learning Outcomes

- Conduct data acquisition using a varied set of techniques structured and unstructured datasets; including raw data files, SQL databases, online repositories, and programmatically through web scraping and APIs.
- Apply preprocessing strategies to complex and dynamic datasets using industry-standard toolkits and machine learning algorithms to extract features, reduce dimensionality, remove errors, inconsistencies, and missing values.
- Differentiate between machine learning approaches such as classification, regression, clustering, and neural networks for predictive analytics and pattern recognition.
- Evaluate the predictive power of the different statistical and machine learning methods to solve real-world data science problems.
- Develop storytelling and visualization techniques to effectively communicate (exploratory) or persuade (explanatory) findings to a specific audience.
- Critically asses ethical considerations and challenges related to data collection and analysis.
- Construct a comprehensive data science project from inception to presentation, integrating the various techniques and tools learned throughout the course.

DSCI530. STATISTICAL METHODS I. 3.0 Semester Hrs.
Introduction to probability, random variables, and discrete and continuous probability models. Elementary simulation, data summarization and analysis using the R Data Analysis Environment. Confidence intervals and hypothesis testing for means and variances. Chi square tests.

Distribution-free techniques and regression analysis. Students are expected to have knowledge of probability covered in MATH334 or an equivalent course. Prerequisite: MATH334 or equivalent.

DSCI560. INTRODUCTION TO KEY STATISTICAL LEARNING METHODS I. 3.0 Semester Hrs.
Part one of a two-course series introducing statistical learning methods with a focus on conceptual understanding and practical applications. Methods covered will include Introduction to Statistical Learning, Linear Regression, Classification, Resampling Methods, Basis Expansions, Regularization, Model Assessment and Selection. Prerequisite: DSCI530 or MATH530.

DSCI561. INTRODUCTION TO KEY STATISTICAL LEARNING METHODS II. 3.0 Semester Hrs.
Equivalent with MATH561,
Part two of a two course series introducing statistical learning methods with a focus on conceptual understanding and practical applications. Methods covered will include Non-linear Models, Tree-based Methods, Support Vector Machines, Neural Networks, Unsupervised Learning. Prerequisite: DSCI560 or MATH560.

DSCI570. INTRODUCTION TO MACHINE LEARNING. 3.0 Semester Hrs.
(I, II) The goal of machine learning is to build computer systems that improve automatically with experience, which has been successfully applied to a variety of application areas, including, for example, gene discovery, financial forecasting, and credit card fraud detection. This introductory course will study both the theoretical properties of machine learning algorithms and their practical applications. Students will have an opportunity to experiment with machine learning techniques and apply them to a selected problem in the context of term projects. Graduate students must complete a more challenging project that utilizes complex machine learning algorithms, requiring a deeper understanding of machine learning approaches and critical thinking. Prerequisite: DSCI503.
Course Learning Outcomes

- Apply supervised, unsupervised, reinforcement machine learning models and deep learning models to solve problems in areas such as prediction, recognition and classification.
- Explore and develop with various tools, techniques and libraries in Python for data processing, feature extraction, visualization, validation and evaluation.
- Create data visualization tools, techniques, and libraries in Python to visualize high dimensional or complex data for stakeholders.
- Determine ethical implications through interpretability of big data and results from the application of various machine learning models.
- Design and develop a machine learning product that solves their chosen real-world challenge.
- Create a video presentation that succinctly outlines the problem, solutions, conclusions, and lessons learned regarding product development for the stakeholders.

DSCI575. ADVANCED MACHINE LEARNING. 3.0 Semester Hrs.
The goal of machine learning research is to build computer systems that learn from experience and that adapt to their environments. Machine learning systems do not have to be programmed by humans to solve a problem; instead, they essentially program themselves based on examples of how they should behave, or based on trial and error experience trying to solve the problem. This course will focus on the methods that have proven valuable and successful in practical applications. The course will also contrast the various methods, with the aim of explaining the situations in which each is most appropriate. Prerequisite: DSCI570.

DSCI576. DEEP LEARNING. 3.0 Semester Hrs.
Machine Learning is a key component of Artificial Intelligence that allows computers to generate accurate outcomes without being explicitly programmed. Over the past decade, machine learning has powered speech recognition, augmented reality, self-driving vehicles, and biometric authentications. The purpose of this course is to bridge general machine learning concepts to real world applications. This course provides a broad introduction to machine learning (supervised learning, unsupervised learning, feature learning), deep learning, computer vision, natural language processing, and reinforcement learning. During the course, students will design a project that applies concepts from class to several different real-world problems and challenges and create applications to solve them. Prerequisite: CSCI470.
Course Learning Outcomes

- Design deep learning models to address supervised and unsupervised learning tasks implemented in Python that meet evaluation criteria such as accuracy, F1 score, and mean-square error.
- Apply state-of-the-art deep learning architectures for computer vision tasks involving object detection, image generation, and self-driving vehicles.
- Develop deep learning models for natural language processing tasks using pre-trained transformers.
- Utilize Markov Decision Process, Monte Carlo Method, and deep reinforcement learning techniques to solve sequential decision-making problems.
- Evaluate ethical challenges in deep learning models, including bias, fairness, transparency, and interpretability.
- Demonstrate clear communication of deep learning results and insights to both technical and non-technical audiences.

DSCI598. SPECIAL TOPICS. 0-6 Semester Hr.

DSCI598. SPECIAL TOPICS. 1-6 Semester Hr.

DSCI598. SPECIAL TOPICS. 1-6 Semester Hr.

DSCI598. SPECIAL TOPICSE. 1-6 Semester Hr.

DSCI598. DATA SCIENCE. 1-6 Semester Hr.

DSCI599. INDEPENDENT STUDY. 0.5-6 Semester Hr.

DSCI599. INDEPENDENT STUDY. 0.5-6 Semester Hr.

Professors

Soutir Bandyopadhyay, Applied Mathematics and Statistics

Dorit Hammerling, Applied Mathematics & Statistics

Teaching Professor

Wendy Fisher, Computer Science

Teaching Assistant Professors

Nathan Lenssen, Applied Mathematics and Statistics

Zibo Wang, Computer Science

Research Associate Professor

Zane Jobe, Geology and Geological Engineering

Professor Emeritus

Douglas Nychka, Applied Mathematics & Statistics

FEA Professional

Degrees Offered

- Graduate Certificate – FEA Professional

Program Description

The Graduate Certificate – FEA Professional is a fully online graduate-level certificate program that teaches advanced skills in finite-element analysis for structural applications. The program has been designed to train recent graduates or midcareer professionals with at least a BS in engineering, computer science, or applied engineering physics who are interested in careers and/or opportunities in design, product development, or applied research. The program leverages industry-leading software to empower students with the skills and experience to drive innovation in their chosen field. Our courses help students build a foundation of practical knowledge focused on key fundamentals of applied computational mechanics complemented by the perfect balance of theoretical background. The fundamentals learned here may be applicable across a broad range of industries. Upon completion of the program, graduates will have the skills to: a) earn software-specific endorsements/certifications for industry-leading products such as Abaqus, SolidWorks Simulation, ANSYS, b) drive innovation through the effective application of simulation tools for ideation and design verification, c) leverage simulation to reduce physical testing in new product development (NPD) and reduce time to market, d) exploit parametric simulation, DOE, and optimization to reveal more and better R&D solutions, e) identify CTQ's in your development or applied research projects and quantify impact on all relevant outcome metrics, f) execute, review, and manage simulation strategies for your NPD or applied research pipeline, and g) spec software tools and training for your team.

Primary Contact

FEA Professional Program
https://mechanical.mines.edu/feapro/

Associate Professor and Program Director

Anthony J. Petrella, Mechanical Engineering

Professor of Practice

Brian Gockel, Mechanical Engineering

Adjunct Professor

Catherine M. Amodeo, PhD, Ford Motor Company

Program Requirements

The Graduate Certificate – FEA Professional requires a set of three core courses (Table 1) and one elective chosen from selected relevant online courses (Table 2). Elective options will continue to expand as the program matures.

The student must complete the following three core courses.

FEGN525	ADVANCED FEA THEORY & PRACTICE	3.0
FEGN526	STATIC AND DYNAMIC APPLICATIONS IN FEA	3.0
FEGN527	NONLINEAR APPLICATIONS IN FEA	3.0

The student must complete one of the following elective courses (3 semester hrs).

FEGN528	FEA FOR ADVANCED DESIGN APPLICATIONS	3.0
AMFG521	DESIGN FOR ADDITIVE MANUFACTURING	3.0

Transfer Credits

Transfer credits are not currently accepted to satisfy requirements for the Graduate Certificate – FEA Professional.

Prerequesites

A baccalaureate degree in engineering, computer science, a physical science, or mathematics is required.

Courses

FEGN525. ADVANCED FEA THEORY + PRACTICE. 3.0 Semester Hrs.
This course examines the theory and practice of finite element analysis. Direct methods of deriving the FEA governing equations are addressed as well as more advanced techniques based on virtual work and variational methods. Common 1D, 2D, and 3D element formulations are derived, and key limitations examined. Matlab is used extensively to build intuition for FEA solution methods and students will create their own 2D FEA code by the end of the course. The commercial FEA software Abaqus is introduced with hands-on examples and Matlab solutions are compared to Abaqus for model validation.
Course Learning Outcomes

- Define DOF.
- Recall three different approaches for developing governing equations in FEA and list typical applications for each.
- Apply FEA governing equations to solve2D structural analysis by hand using symbolic math in Matlab.
- Explain and execute a mesh convergence study.
- Define the isoparametric element formulation and use shape functions to derive isoparametric elements for 2D and 3D applications.
- Recall numbers and locations of integration points for different element types.
- List and explain limitations of common 2D and 3D elements.

FEGN526. STATIC AND DYNAMIC APPLICATIONS IN FEA. 3.0 Semester Hrs.
This course emphasizes proficiency with commercial FEA software for solution of practical static, quasistatic, and dynamic structural problems. Common 1D, 2D, and 3D elements are examined in the context of linear solution techniques. Students will explore efficient methods for model construction and solution with commercial tools (the Abaqus FEA software). Emphasis will also be placed on verification, validation, and reporting standards for effective application of FEA software tools. Online course. Prerequisite: FEGN525.
Course Learning Outcomes

- Explain the difference between implicit and explicit solvers for static, quasi-static, and dynamic analyses.